

Violet Yinuo Han

Ph.D. Student, Human-Computer Interaction Institute, Carnegie Mellon University

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Research Interests

INTERFACES AND INTERACTIONS WITH EMBODIED AGENTS • HUMAN-ROBOT INTERACTION • PERSONAL ROBOTS • USER MODELING

My research builds toward a future of diverse personal robots that understand people and fit into everyday life.

I develop methods that let robots *learn what matters to each person* from everyday activity with minimal user burden, combining the open-world knowledge of foundation models with the structured, uncertainty-aware reasoning of probabilistic models. I explore the *forms* personal robots can take and how form shapes what they can do and how people work with them, and I design *interfaces that build on the objects and spaces* people already have.

Education

Ph.D. Student , Carnegie Mellon University, School of Computer Science Human-Computer Interaction Institute, advised by Prof. Alexandra Ion	Started August 2023
M.S. , Carnegie Mellon University <i>Thesis</i> : Building a Bidirectional Bridge Between the Digital and Physical Worlds	Graduated May 2023
B.F.A. , University of Michigan <i>Thesis</i> : 86 Billions: Experiencing the Human Brain From a Neuron's Perspective	Graduated May 2020

Publications

I have published 10 peer-reviewed publications, 8 as first author, and received 2 best-paper/invited-extension recognitions.

- [1] Multimodal Intent Inference via LLM-Augmented Dynamic Bayesian Networks**
Violet Yinuo Han, Hyunsung Cho, Tianyi Wang, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, Tanya R. Jonker, Alexandra Ion
Proceedings of ACM Transactions on Interactive Intelligent Systems (Highlights of IUI 2025).
- [2] Priors of U: Dynamic Preference Learning for Embodied Interaction**
Violet Yinuo Han, Alexandra Ion
Proceedings of ACM UIST'26, Detroit, MI. Nov. 2–5, 2026 (21.4% acceptance rate)
- [3] Objects with Arms: How Robot Embodiment Shapes Perception in Physical Collaboration**
Violet Yinuo Han, Zirui Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion
Proceedings of IEEE RO-MAN'26, Kitakyushu, Japan. Aug. 24–28, 2026
- [4] Towards Unobtrusive Physical AI: Augmenting Everyday Objects with Intelligence and Robotic Movement for Proactive Assistance**
Violet Yinuo Han, Jesse T. Gonzalez, Christina Yang, Zhiruo Wang, Scott E. Hudson, Alexandra Ion
Proceedings of ACM UIST'25, Busan, Korea. Sept. 28 – Oct. 1, 2025 (22.2% acceptance rate)
- [5] Transforming Everyday Objects into Dynamic Interfaces using Smart Flat-Foldable Structures**
Violet Yinuo Han, Amber Yinglei Chen, Mason Zadan, Jesse T. Gonzalez, Dinesh K. Patel, Wendy Fangwen Yu, Carmel Majidi, Alexandra Ion
Proceedings of ACM UIST'25, Busan, Korea. Sept. 28 – Oct. 1, 2025 (22.2% acceptance rate)
- [6] A Dynamic Bayesian Network Based Framework for Multimodal Context-Aware Interactions**
Violet Yinuo Han, Tianyi Wang, Hyunsung Cho, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, Alexandra Ion, Tanya R. Jonker
Proceedings of ACM IUI'25, Cagliari, Italy. March 24–27, 2025 (25% acceptance rate) **Invited Journal Extension**

- [7] **Robotic Metamaterials: A Modular System for Hands-On Configuration of Ad-Hoc Dynamic Applications**
Zhitong Cui, Shuhong Wang, **Violet Yinuo Han**, Tucker Rae-Grant, Willa Yunqi Yang, Alan Zhu, Scott E. Hudson, Alexandra Ion
Proceedings of ACM CHI'24, Honolulu, HI. May 11–16, 2024 (26.4% acceptance rate)
- [8] **Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices**
Violet Yinuo Han, Abena Boadi-Agyemang, Yuyu Lin, David Lindlbauer, Alexandra Ion
Proceedings of ACM UIST'23, San Francisco, CA. Oct. 29 – Nov. 1, 2023 (21% acceptance rate)
- [9] **BlendMR: A Computational Method to Create Ambient Mixed Reality Interfaces**
Violet Yinuo Han, Hyunsung Cho, Kiyosu Maeda, Alexandra Ion, David Lindlbauer
Proceedings of ACM ISS'23, Pittsburgh, PA. Nov. 5–8, 2023 (24–28% acceptance rate) **Best Paper Award, Top 1**
- [10] **Permeable Thermistor Temperature Sensors Based on Porous Melamine Foam**
Hugo de Souza Oliveira, Niloofar Saeedzadeh Khaanghah, **Violet Yinuo Han**, Alejandro Carrasco-Pena, Alexandra Ion, Michael Haller, Giuseppe Cantarella, Niko Münzenrieder
IEEE Sensors Letters, Vol. 7, No. 5, pp. 1–4, Art. no. 2500904, May 2023

Awards and Grants

Qualcomm Innovation Fellowship, \$100k, with Weiwei Sun	2026
Special Recognition for Outstanding Review, ACM UIST	2025, 2026
Best Paper Award, Top 1, ACM ISS'23 for <i>BlendMR</i> [8]	2023
Snap Creative Challenge Award, \$10k, with Prof. David Lindlbauer and Dr. Yukang Yan	2022
Carnegie Mellon University SoA Commendation	2022
Carnegie Mellon University SoA Merit Scholarship, \$10k	2021
University of Michigan University Honors	2018, 2020

Professional Experience

Research Scientist Intern, Meta Reality Labs	May – Sept. 2024
Proposed a dynamic Bayesian network framework for multimodal context-aware interaction. Published at IUI'25 [5] and invited for journal extension.	
System Experience Design Intern, OnePlus	May – July 2021
Worked with designers and engineers to maintain and update OnePlus OxygenOS.	
Interaction Design Intern, WhaleDynamic	Feb. – May 2021
Designed interactions for an autonomous delivery vehicle and presented a prototype at the Shanghai International Automobile Industry Exhibition.	

Invited Talks

CMU Robotic Caregiving and Human Interaction Lab , hosted by Prof. Zackory Erickson	April 2026
The Ellis School , hosted by Prof. Tai Sing Lee	April 2025

Research Demos

IEEE/RSJ IROS , CMU and Fujitsu booths, Towards Unobtrusive Physical AI [3]	Pittsburgh, 2026
IEEE/RSJ IROS , Robotics Innovation Center tour, Towards Unobtrusive Physical AI [3]	Pittsburgh, 2026
CMU Robotics Innovation Center Opening , Towards Unobtrusive Physical AI [3]	Pittsburgh, 2026
CMU XRTC Symposium , Towards Unobtrusive Physical AI [3]	Pittsburgh, 2025
ACM UIST , Towards Unobtrusive Physical AI [3]	Busan, Korea, 2025
CMU XRTC Symposium , Parametric Haptics [7]	Pittsburgh, 2024

Media Coverage

ComputerWorld , “Why ambient robots beat humanoid robots” [3]	2026
CNET , “Robot Coffee Cups? Self-Driving Trivets? AI Researchers Made It Happen” [3]	2025
CMU News , “CMU Researchers Use AI To Turn Everyday Objects Into Proactive Assistants” [3]	2025
CMU SCS News , “This Stapler Knows When You Need It” [3]	2025
TechXplore , “A stapler that knows when you need it: Using AI to turn everyday objects into proactive assistants” [3]	2025
The Brighter Side of News , “AI revolution: Researchers teach everyday objects to sense, think, and move” [3]	2025

Teaching Experience

Teaching Assistant , Carnegie Mellon University	Aug. – Dec. 2026
<i>Human-AI Interaction</i> , Prof. Sherry Wu and Prof. Sauvik Das	
Guest Lecturer , Carnegie Mellon University	April 2025
<i>Computational Methods for Interactive Systems</i> (05-499/899), Prof. Alexandra Ion. Lecture on using dynamic Bayesian networks to model interactions for building interactive systems.	
Teaching Assistant , Carnegie Mellon University	Aug. – Dec. 2022
<i>Computational Perception</i> (15-387/86-375/675), Prof. Tai Sing Lee. Hosted weekly office hours, created and graded assignments, and wrote lecture notes to support student learning.	

Mentoring

Danny Linitz , Design at CMU, Undergraduate Research Assistant	Fall 2026
Selina Lee , Design at CMU, Undergraduate Research Assistant	Fall 2026
Dian Zhu , Mathematics at CMU, Undergraduate Research Assistant	Fall 2025
Narayan Ashanahalli , Architecture at CMU, Masters Research Assistant	Fall 2025
Aiden Li , Computer Science at UCL, Undergraduate Research Assistant	Summer 2025
Christina Yang , Design and HCI at CMU, Masters Research Assistant	Fall 2024
Anagha Srinivasan , Computer Science at CMU, Undergraduate Research Assistant	Spring 2024
Ziyu Li , ECE at CMU, Undergraduate Research Assistant	Fall 2023
Wendy Yu , Design and HCI at CMU, Undergraduate Research Assistant	Fall 2023, Spring 2024

Academic Service

Poster Area Chair , ACM UIST	2026
Reviewer , PAMHCI (ISS)	2026
Reviewer , ACM UIST	2024, 2025, 2026
Reviewer , ACM CHI	2024, 2025, 2026
Reviewer , ACM CHI Late-Breaking Work	2023, 2024
Assistant to Session Chair , ACM CHI	2024
Student Volunteer , ACM UIST	2022